

ICP 02. Differential Diagnosis and Diagnostic Traps

Companion reading handout for simultaneous listening.

Audio course on intrahepatic cholestasis of pregnancy. Episode two. The differential diagnosis traps.

Source Base

This episode is based on Gabbe chapters on hepatic disease, skin disease, gastrointestinal disease, preeclampsia, renal disease, and the Israeli position papers on cholestasis, hypertension, hepatitis C, and nausea and vomiting of pregnancy.

The goal of this episode is safety.

Intrahepatic cholestasis of pregnancy, or ICP, is important because of fetal risk. But the mimics of ICP are important because some of them are maternal emergencies. A resident must be able to say, this is probably ICP, but also, this is not allowed to be ICP until I have excluded something more dangerous.

Pruritus Without Primary Rash

Let us begin with pruritus.

Pruritus in pregnancy splits into two practical worlds. Pruritus without a primary rash, and pruritus with a primary rash.

In pruritus without a primary rash, systemic disease rises on the list. ICP is one of the most important causes. But it is not alone. Think of viral hepatitis, biliary obstruction, drug-induced cholestasis, renal disease, thyroid disease, and rarely malignancy such as lymphoma.

The patient may have only excoriations. That does not count as a primary eruption. Excoriations are the footprints of scratching. They tell you the patient is itchy, not why.

Pruritus With Primary Rash

Now pruritus with a primary rash is a different diagnostic universe.

Polymorphic eruption of pregnancy often starts in abdominal striae and usually spares the umbilicus. It is very itchy, but it is a skin disease, not a bile-acid fetal-risk disease.

It is also associated with multiple gestation, probably because abdominal wall distension and hormonal burden are part of the biology.

Pemphigoid gestationis is rarer, but more serious. It often starts around the umbilicus with urticarial lesions that can become vesicular or bullous. It can flare around delivery. It has autoimmune features, and the diagnosis may require immunofluorescence.

Atopic eruption of pregnancy tends to appear earlier, often before the third trimester, and has eczematous or papular features. Prurigo has grouped excoriated papules, often on extensor surfaces. Pruritic folliculitis has follicular papules and pustules.

The audio memory hook is straightforward.

ICP is itch with no primary rash. If you see the rash before the scratch marks, slow down.

Liver Tests

Now let us move from skin to liver tests.

The dangerous mistake is to see elevated AST or ALT and say, this must be ICP, because the patient is pregnant and itchy. That is not enough.

Gabbe's hepatic chapter places ICP in a broader third-trimester differential with preeclampsia, HELLP syndrome, acute fatty liver of pregnancy, viral hepatitis, and biliary disease.

Preeclampsia and HELLP

Start with preeclampsia and HELLP.

Preeclampsia is not a liver disease, but it can injure the liver. The Israeli hypertension position paper defines severe features to include elevated liver enzymes, and HELLP means hemolysis, elevated liver enzymes, and low platelets.

So ask the simple questions. Is the blood pressure elevated? Is there proteinuria or another severe feature? Are the platelets falling? Is there hemolysis? Is there renal dysfunction? Is there headache, visual disturbance, pulmonary edema, or epigastric pain?

If those are present, ICP may coexist, but it cannot explain the whole picture.

Here is the difference in clinical tone. ICP usually itches. HELLP often hurts, pressures rise, platelets fall, and the mother can deteriorate.

Acute Fatty Liver of Pregnancy

Now acute fatty liver of pregnancy, or AFLP.

AFLP is rare, but it is one of the diagnoses you cannot miss. It is a systemic illness. The patient may have nausea, vomiting, malaise, fatigue, right upper quadrant pain, hypoglycemia, coagulopathy, renal dysfunction, or encephalopathy. Delivery after maternal stabilization is central.

If the mother looks ill, if glucose is low, if coagulation is abnormal, if renal function is worsening, do not hide behind ICP. ICP is usually not a maternal multi-organ failure syndrome.

Viral Hepatitis

Now viral hepatitis.

Viral hepatitis can present with malaise, anorexia, nausea, jaundice, abdominal discomfort, and markedly elevated transaminases. Herpes hepatitis in pregnancy is rare but dangerous, and Gabbe emphasizes right upper quadrant pain, fever or prodrome, and very high aminotransferases. Hepatitis C and other chronic viral diseases also matter because background liver disease changes how confidently we interpret new cholestasis.

In ICP, transaminases may be severalfold elevated. But extremely high values, systemic viral symptoms, exposure risk, or persistent abnormalities postpartum should push you toward hepatitis evaluation.

Gallbladder and Biliary Obstruction

Now gallbladder and biliary obstruction.

Pregnancy itself slows gallbladder emptying and makes bile more lithogenic. That means gallstones and sludge become more likely. Biliary obstruction tends to bring pain, especially right upper quadrant pain or biliary colic. Fever and jaundice suggest infection or obstruction and require urgent attention.

ICP can cause pruritus and mild direct bilirubin elevation. But it should not cause a classic obstructive pain story with abnormal imaging. If pain dominates, image the biliary tract.

Hyperemesis Gravidarum

Now hyperemesis gravidarum.

This is usually an early pregnancy disease. The Israeli nausea and vomiting position paper defines hyperemesis by persistent vomiting not explained by other disorders,

with weight loss over five percent, dehydration, and ketonuria. Liver dysfunction can occur in hyperemesis.

So timing matters. A woman at nine weeks with vomiting, ketonuria, and mild transaminitis is not the classic ICP patient. A woman at thirty-four weeks with nocturnal palm and sole pruritus is.

Drug-Induced Cholestasis

Now drug-induced cholestasis.

This is easy to forget. Always ask what changed. Antibiotics, antiepileptics, supplements, herbal products, hormones, and other medications can injure the liver or cause cholestasis. A temporal relationship to a new drug is a diagnostic clue.

Renal and Thyroid Disease

Now renal and thyroid disease.

Gabbe's skin chapter reminds us that pruritus without eruption can reflect renal or thyroid disease. Uremia can itch. Thyroid disease can itch. These are not the commonest explanations in a classic third-trimester ICP story, but they matter when bile acids are normal and symptoms do not fit.

Bedside Sorting System

Let us now build a practical bedside sorting system.

Ask first, is there a primary rash?

If yes, decide whether this is a pregnancy dermatosis, allergic reaction, infection, infestation, drug eruption, or another skin disease. ICP can coexist with skin disease, but the rash deserves its own diagnosis.

Ask second, is the mother systemically ill?

Fever, severe vomiting, malaise, hypoglycemia, coagulopathy, encephalopathy, renal dysfunction, or severe abdominal pain should make you step away from uncomplicated ICP.

Ask third, is there hypertensive disease?

Blood pressure, platelets, creatinine, urine protein, symptoms, and hemolysis decide whether preeclampsia or HELLP is in the room.

Ask fourth, does the timing make sense?

ICP usually appears late. Earlier disease can happen, but a first-trimester presentation should make you ask harder questions.

Ask fifth, do the labs behave like ICP?

Bile acids should be central. AST and ALT may be elevated. Bilirubin may be mildly direct. Alkaline phosphatase is hard to interpret in pregnancy. Marked synthetic dysfunction, severe thrombocytopenia, hypoglycemia, or extreme systemic derangement is not classic ICP.

Case Contrast

Now let us put it into a short case contrast.

Case one. Thirty-five weeks, palms and soles itch at night, no rash except excoriations, bile acids thirty-two, mild ALT elevation, normal blood pressure, normal platelets. This is ICP until something else appears.

Case two. Thirty-five weeks, itching, epigastric pain, blood pressure severe range, platelets ninety thousand, AST elevated. This is hypertensive disease with severe features, with or without ICP.

Case three. Thirty-five weeks, itch and jaundice, fever, right upper quadrant pain, high bilirubin, dilated ducts on ultrasound. This is biliary obstruction or cholangitis territory.

Case four. Thirty-five weeks, itch, tense bullae, periumbilical involvement. Think pemphigoid gestationis.

The resident-level point is not to memorize every differential. It is to avoid premature closure.

Before we end, here is the synthesis.

Summary

First, pruritus without a primary rash is where ICP lives, but systemic causes also live there.

Second, pruritus with a primary rash deserves dermatologic thinking. The morphology and distribution are not cosmetic details. They are diagnostic data.

Third, abnormal liver tests in late pregnancy require active exclusion of preeclampsia, HELLP, AFLP, viral hepatitis, biliary obstruction, hyperemesis-related disease, and drug injury.

Fourth, ICP is usually uncomfortable for the mother. Several mimics are dangerous for the mother.

That is the key safety sentence.

ICP itches. Its mimics can kill.

End of episode two.